Subject: Computer Studies

# TOPIC ONE: (COMMUNICATION SYSTEM)

## Information and Communication Technology (ICT)

Information technology (IT) is the study or use of systems (especially computers and telecommunications) for storing, retrieving, and sending information.
ICT is an umbrella term that includes any communication device or application, encompassing: radio, television, cellular phones, computer and network hardware and software, satellite systems and so on, as well as the various services and applications associated with them.

**Types of ICT**

1. **Broadcasting:** Broadcasting is the distribution of audio and/or video content or other messages to a dispersed audience via any electronic mass communications medium, but typically one using the electromagnetic spectrum (radio waves), in a one-to-many model.
2. **Telecommunication:** It is the transmission of messages over significant distance(s) for the purpose of communication. A basic telecommunication system consists of three primary units that are always present in some form:

- **Transmitter:** A transmitter takes information and converts it to a signal.
- **Transmission medium:** this is the physical channel that carries the signal.
- **Receiver:** A receiver takes the signal from the channel and converts it back to usable information.

3. **Data Networks:** A network is a connection of electronic devices especially the computers for the purpose of communication and sharing of resources. A data network is an electronic communication process that allows for the orderly transmission and reception of data. Such as letters, spreadsheets and other types of document.
4. **Information Systems:** the term information system is used to refer to the interaction between people, processes, data and technology. In this sense, the term is used to refer not only the information and communication technology (ICT) organization uses but also to the way in which people interact with technology in support of business processes. Examples of information systems are Management Information System (MIS), Decision Support System (DSS), Geographical Information System (GIS) etc.
5. **Satellite communication:** A satellite is a type of space craft that is launched into space and equipped with sensors to record and transmit information. Satellite TV is a broadcasting service that allows subscribers to receive television signals through a dish-shaped receiver unit.

**Types of broadcasting**

1. **Telephone broadcasting:** the earliest form of electronic broadcasting that uses telephone as the medium of broadcast.

2. **Radio broadcasting:** involves the transmission of audio signal through the air as radio waves from a transmitter, picked up by an antenna and sent to the receiver (radio).

3. **Television broadcasting:** involves the transmission of video signals in addition to audio signals from a transmitter to an antenna and finally to the receiving television.

4. **Satellite TV systems:** are meant for direct-to-home broadcast programming. It offers a mix of traditional radio or television station broadcast programming or both, with dedicated satellite radio programming. Examples include cable television station such as CNN, BBC, SuperSport etc.

5. **Webcasting:** It is the broadcasting of live or delayed audio/video information using the internet.

**Types of Telecommunication Systems**

1. **Public Switched Telephone Network** (PSTN) – land line

   PSTN also refers to as Plain Old Telephone Service (POTS), is a worldwide net of telephone lines using fiber optic cables, microwave transmission links, cellular networks and communication satellite connected by switching centers, which allows any telephone in the world to communicate with any other.

2. **Mobile Phone System** (GSM)

   GSM stands for Global System for Mobile communication. It is the most popular standard for mobile telephony systems in the world. GSM is considered a “second generation (2G)” mobile phone system. GSM uses a SIM (Subscriber Identity Module) card. The SIM card is tied to your cell phone service carrier’s network rather than to the handset itself. Virtually all mobile phones are 3G compliant while some countries have also launched the 4G network.

3. **Circuit Switched Packet Telephone Systems** (CSPT)

   When you hear the term circuit switching, think phone call. A circuit switching network establishes a circuit (or channel) between a sender and a receiver before they can communicate, as if the sender and the receiver were physically connected with an electric circuit. If communication is taking place in a dedicated channel, the channel remains unavailable to other users.

4. **Satellite Telephone System:** A satellite telephone connects to the orbiting satellites instead of terrestrial cell sites like the MTN or GLO mast. Depending on the architecture of a particular system, coverage may include the entire earth or only specific regions.

5. **Fixed wireless telephone system:** Fixed wireless is the operation of wireless devices or systems used to connect two fixed location (e.g. building) with a radio or other wireless link, such as Laser Bridge. The purpose of a fixed

wireless telephone link is to enable telecommunications between the two sites or buildings.

**Types of Data networks.**

1. **Local Area Network** (LAN): A LAN is a computer network that connects devices such as computers, hubs, switches etc over a relatively short distance. This kind of network is confined to a single building such as an office building, hospitals, schools or home.

2. **Metropolitan Area Network** (MAN): It refers to a network that exist within a single city or a metropolitan area. A MAN is larger than a LAN but smaller than a WAN. If we had two different building within a city that were connected together , it would be considered a MAN.

3. **Wide Area Network** (WAN): As the name implies, a WAN covers a large geographical locations and is typically made up of multiple LANs. A network device called a Router connects this LANs together to form a WAN. A typical example of a WAN is the internet.

4**. Internet:** The internet literally means internetwork or international network. An internetwork is a type of WAN that connects bunch of networks. The Internet is a global system of interconnected computer networks that use the standard Internet Protocol Suite (TCP/IP) to link several billion devices together.

**Types of information systems**

1. **Data processing system**: This is a system that processes data, which has been captured and encoded in a format recognized by the data processing system or has been created and stored by another unit of an information system.

2. **Global positioning system** (GPS): Is a space-based global navigation satellite that provides reliable location and time information in all weather and at all times and anywhere on or near the earth when and where there is an unobstructed line of sight to four or more GPS satellites. The GPS is made up of 3 parts:

i. The satellites orbiting the Earth called the space segment;

ii. The control and monitoring stations on earth called the control station;

iii. The GPS receiver.

**GPS Applications**

1. **Navigation:** GPS allows soldiers to find their targets

2. **Map making:** Both civilian and military cartographers use GPS extensively.

3. **Disaster relief/emergency services**:they depend upon GPS for location and timing capabilities

4. **Geofencing:** Vehicle or any other type of tracking systems use GPS to locate a vehicle etc.

5 **Surveying:** surveyors use absolute locations to make maps and determine property boundary.

6. **Search and rescue :** downed pilots can be located faster if their position is known.

7. **Cellular telephony:** GPS receivers are now integrated in many mobile phones

8. **Missile and projectile guidance:** GPS allows accurate targeting of various military weapons.